Assignment – 1.14

Introduction to IP Addressing Theory

Q - 1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

➢ **Differences Between IPv4 and IPv6:**

IPv4	IPv6
1. Uses a 32-bit address.	1. Uses a 128-bit address.
2. Uses numbers separated by dots.	2. Uses numbers and letters separated by colons.
3. Example: 192.168.1.1	3. Example: 2001:db8::1
4. Provides fewer IP addresses.	4. Provides a huge number of IP addresses.
5. Uses a subnet mask.	5. Uses prefix length, such as /64.

<u>Q - 2.</u> Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

- **<u>Static IPv4 Address Assignment:</u>**

  Network: **192.168.10.0/24**

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.2	255.255.255.0	192.168.10.1
Phone	192.168.10.3	255.255.255.0	192.168.10.1
Smart TV	192.168.10.4	255.255.255.0	192.168.10.1
Printer	192.168.10.5	255.255.255.0	192.168.10.1

Q - 3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

- **Number of subnets:**

  $2^3 = 8$subnets

- **Usable hosts per subnet:**

  $2^5 - 2 = 30$hosts

- **First three subnet ranges:**

Subnet	Network Address	Usable Host IPs	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

- **Final Answer:**

  - Number of subnets: **8**
  - Usable hosts per subnet: **30**
  - Each subnet contains **32 total IP addresses**.

Q - 4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet. Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

- **Step 1: Convert subnet mask to CIDR:**

  255.255.255.0 = /24

- **Step 2: Find the network address:**

  Network address = 10.0.5.0

- **Step 3: Find the broadcast address:**

  Broadcast address = 10.0.5.255

- **Step 4: Find the valid host range:**

  First usable IP: 10.0.5.1

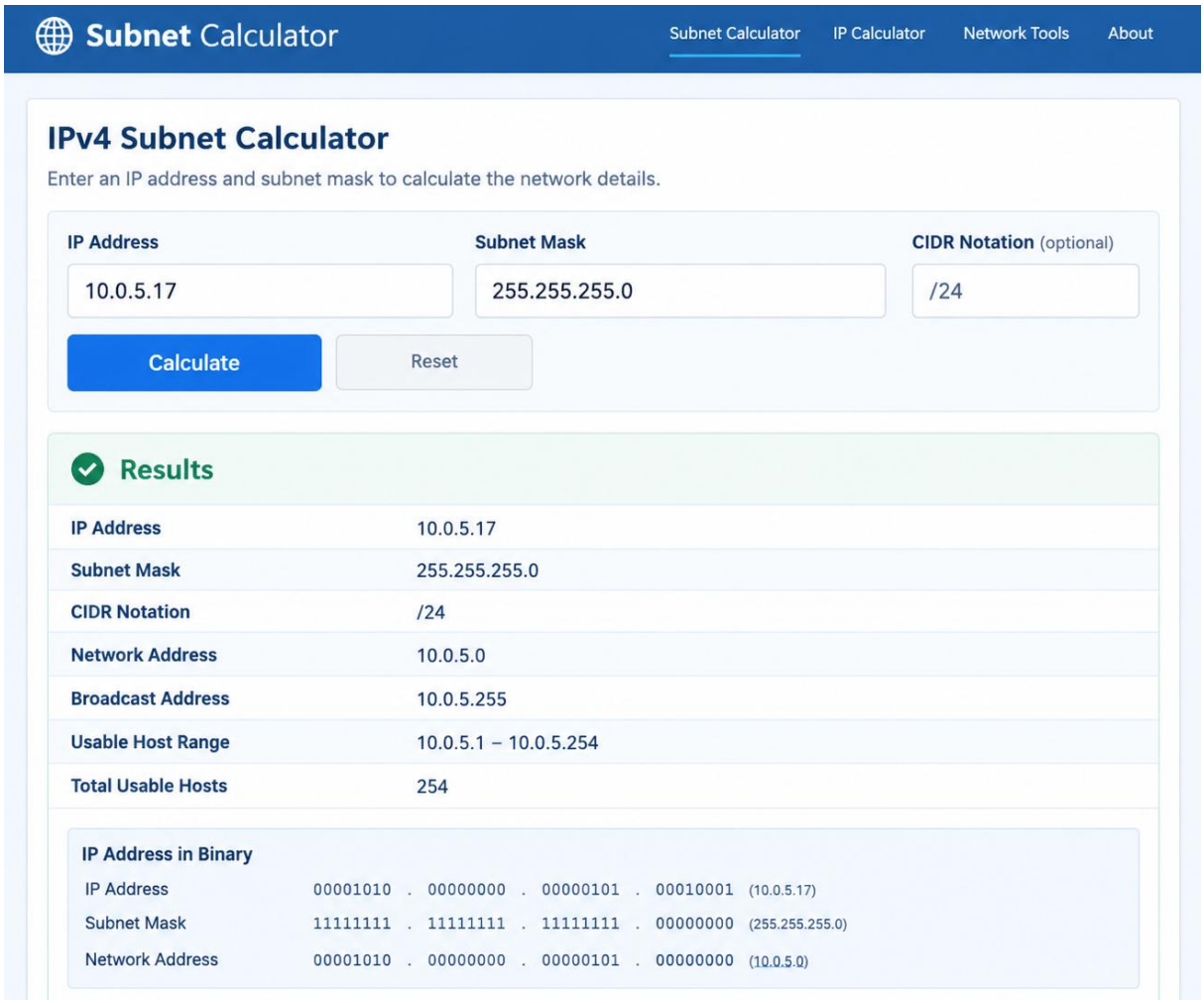
  Last usable IP: 10.0.5.254

- **Final Answer**

Item	Answer
CIDR notation	/24
Network address	10.0.5.0

Item	Answer
Broadcast address	10.0.5.255
Valid host range	10.0.5.1 – 10.0.5.254
Total usable hosts	254

Q - 5. Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.



- **Explanation:** I entered the IP address 10.0.5.17 and subnet mask 255.255.255.0 into the subnet calculator. The results matched my manual answer.